

Module 3

Business Economics

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)(i)

Marginal analysis is a decision-making tool that compares the benefits arising from an economic activity (marginal benefit ("MB")) with its associated costs (marginal cost ("MC")) in order to determine if the activity is worth undertaking.

Answer 1(a)(ii)

The opportunity cost of a particular choice (buying something or undertaking some activity) is the value of the next highest-valued alternative use of that resource.

Opportunity cost captures the marginal cost in the marginal analysis and is measured by the sum of the monetary costs incurred for the choice taken (direct costs) and the monetary value of the benefits provided by the best forgone alternative (indirect costs).

Answer 1(a)(iii)(1)

Decision: To buy or not buy new tickets – Marginal Analysis

Since $MB (= HK\$3,000) > MC (= HK\$2,500)$, the couple should buy two new tickets and watch the concert.

[For reference only:

Decision rule: Buy the new tickets if the benefit of watching the concert (which is the marginal benefit as reflected by the maximum price the couple is willing to pay for the tickets) is greater than the purchase cost of the new tickets (the marginal cost).]

Answer 1(a)(iii)(2)

Decision: To buy or not buy new tickets – Total Cost-benefit Analysis

Since HK\$2,500 has already been paid for the original tickets (which were subsequently lost), the couple should buy the new tickets as the net loss associated with watching the concert ($HK\$3,000 - HK\$5,000$) is less than that of not watching the concert ($HK\$0 - HK\$2,500$).

[For reference only:

Decision rule: Given the costs already incurred (i.e. the sunk cost), buying the new tickets if the benefit of doing so maximises net benefit (if total benefits > total costs) or minimises the net cost (total benefits < total costs).]

Answer 1(b)(i)

Price elasticity of demand is a measure of the percentage change in the quantity demanded ($\% \Delta Q_d$) of a product in relation to a given percentage change in its price ($\% \Delta P$).

Answer 1(b)(ii)

Demand and supply analysis can only provide information on the direction of the change in sales (quantity) when price is changed.

To determine the size of the change in sale (in percentage) and thus revenue increasing, decreasing or remaining unchanged, the price elasticity of demand would be helpful.

Answer 1(b)(iii)

The law of demand implies that the sales of both products will decrease with price increase.

Revenues of high-end chocolate will decrease, as its price elasticity is greater than one (elastic demand).

Revenues of low-end chocolate will remain unchanged, as its price elasticity is equal to one (uninelastic demand).

Answer 2(a)(i)

A technical recession is identified by two consecutive quarters of decrease in real gross domestic product ("GDP").

Answer 2(a)(ii)

Business cycle refers to the recurrent but irregular ups and downs in overall economic activities as measured by real GDP and other macroeconomic indicators such as unemployment and inflation rates.

Answer 2(a)(iii)

In a business cycle, recession refers to the contraction phase/ stage during which a general slowdown in economic activities is observed.

Answer 2(b)(i)

Fiscal policy: The use of the change in government spending and taxation to influence the performance of an economy.

Contractionary fiscal policy is conducted by increasing tax and/ or reducing government spending that will lead to a decrease in aggregate demand and thus economic output.

Answer 2(b)(ii)

Monetary policy: The use of the change in money supply and thus interest rates to influence the performance of an economy.

Expansionary monetary policy is carried out by the central bank by increasing the money supply, cutting interest rates and thus stimulating investment and consumer spending (i.e., aggregate demand, AD).

Answer 2(b)(iii)

By purchasing bonds from financial institutions (mainly banks), the central bank will inject the liquidity/ funds available in the banking system, which in turn increases the money supply and lowers short-term interest rates. This monetary policy tool is known as Open Market Operations ("OMO").

Answer 2(b)(iv)

The other two monetary policy tools that are commonly used to stimulate economic activities include:

- A reduction in the percentage of deposits that banks must hold in their vaults and with the central bank (i.e. required reserve). This will allow banks to make more loans to businesses and consumers, thereby increasing money supply and stimulating overall economic activities.
- A decrease in the interest rate charged by the central bank on (short-term) loans extended to commercial banks (i.e. discount rate). This will decrease the cost of funds and stimulate investment and consumption demands.

Answer 2(c)

False.

Under the linked exchange rate system (a form of fixed exchange rate regime), the Hong Kong dollar is pegged to the US dollar at a fixed exchange rate of HK\$7.8 to one US dollar. By doing so, the territory has virtually lost its control of domestic interest rates, as it has to follow the monetary policy (and thus the interest rate movement) of the US over time.

Answer 3(a)

Descriptive statistics deals with collection, organisation and presentation of data.

Inferential statistics is concerned with deducing the properties of a population based on analysis of the sample data.

Answer 3(b)(i)

Monthly salary (HK\$'000)	No. of employees (f)	Mid point (x)	f.x	f.x ²
11 – 15	19	13	247	3,211
16 – 20	32	18	576	10,368
21 – 25	33	23	759	17,457
26 – 30	28	28	784	21,952
31 – 35	27	33	891	29,403
36 – 40	17	38	646	24,548
41 – 45	14	43	602	25,886
Total	170		4,505	132,825

Answer 3(b)(ii)

$$\text{Mean: } \frac{\sum fx}{n} = \frac{\$4,505}{170} = 26.50 \quad (\text{HK\$26,500})$$

Median:

$$\text{Median is the } \left(\frac{n}{2}\right)^{th} = \left(\frac{170}{2}\right)^{th} = 85^{\text{th}} \text{ ordered value}$$

Median class is: "26 – 30"

$$m = \frac{1}{f_m} \left(\frac{n}{2} - F_{m-1} \right) \times W + L_m = \frac{1}{28} \left(\frac{170}{2} - 84 \right) \times 5 + 25.5 = 25.679 \quad (\text{HK\$25,679})$$

Answer 3(b)(iii)

Standard deviation

$$s = \sqrt{\frac{\sum fx^2 - (\sum fx)^2/n}{n - 1}} = \sqrt{\frac{132,825 - (4,505)^2/170}{170 - 1}} = \sqrt{79.54} = 8.92 \quad (\text{HK\$8,920})$$

Answer 3(b)(iv)

- (1) The mode remains unchanged because the highest frequency remains the same.
- (2) The median remains unchanged because the median is the middle value of the ordered data and is thus not affected by a change in the value of the outlier (the highest salary group in this case).
- (3) The standard deviation would become smaller in value. This is because standard deviation takes each data value into account, and the swapping of a data point from the highest salary group to the second highest salary group will reduce the dispersion of the data.

Answer 4(a)(i)

The term population refers to a collection of data that consists of all values of interest.

A sample is a subset of data drawn from that population.

Answer 4(a)(ii)

Reasons of working with samples instead of populations include:

- Not always possible to study the entire population.
- A properly drawn sample can still be representative of the population.
- It is less costly to collect a selected subset of data of a population.

Answer 4(b)

Sampling error refers to the possibility that the sample may fail to be perfectly representative of the population.

There would be no sampling error if the population could be studied directly (instead of a sample) or the sample drawn from the population is perfectly representative of the population (which is practically impossible).

Answer 4(c)

Random (probability) sampling is a way of selecting a sample of observations from a population to make inferences about the population. Under this method, each member of the population has a certain chance (but not necessarily an equal chance) to be selected.

Probability sampling methods:

- (1) Simple random sampling: Each member of the population has an equal chance of being selected.
- (2) Systematic sampling: Each member of the entire population is first numbered and then sample members are chosen based on a random starting point but with a fixed interval.
- (3) Stratified sampling: Divides a population into groups (strata) that share a characteristic of interest and then draws a random sample from each strata.
- (4) Cluster sampling: Divides a population into groups (clusters) that are representative of the population and then randomly samples clusters to study.

[Other valid methods are acceptable]

Answer 5(a)

A point estimate for a population parameter, such as the population mean, is the (unbiased) single number estimate of that parameter.

A confidence interval, on the other hand, is a range of values for the population parameter with a level of confidence that the range or interval contains the parameter, such as population mean.

Answer 5(b)

$$p = 230/300 = 0.7666, n=300, Z_{0.05} = 1.96$$

95% C.I. of population proportion of employees

$$= (0.7666 - 1.96 \sqrt{\frac{0.7666(1 - 0.7666)}{300}}, 0.7666 + 1.96 \sqrt{\frac{0.7666(1 - 0.7666)}{300}})$$

$$= (0.7666 - 0.0479, 0.7666 + 0.0479)$$

$$= (0.7187, 0.8145)$$

* * * END OF EXAMINATION PAPER * * *